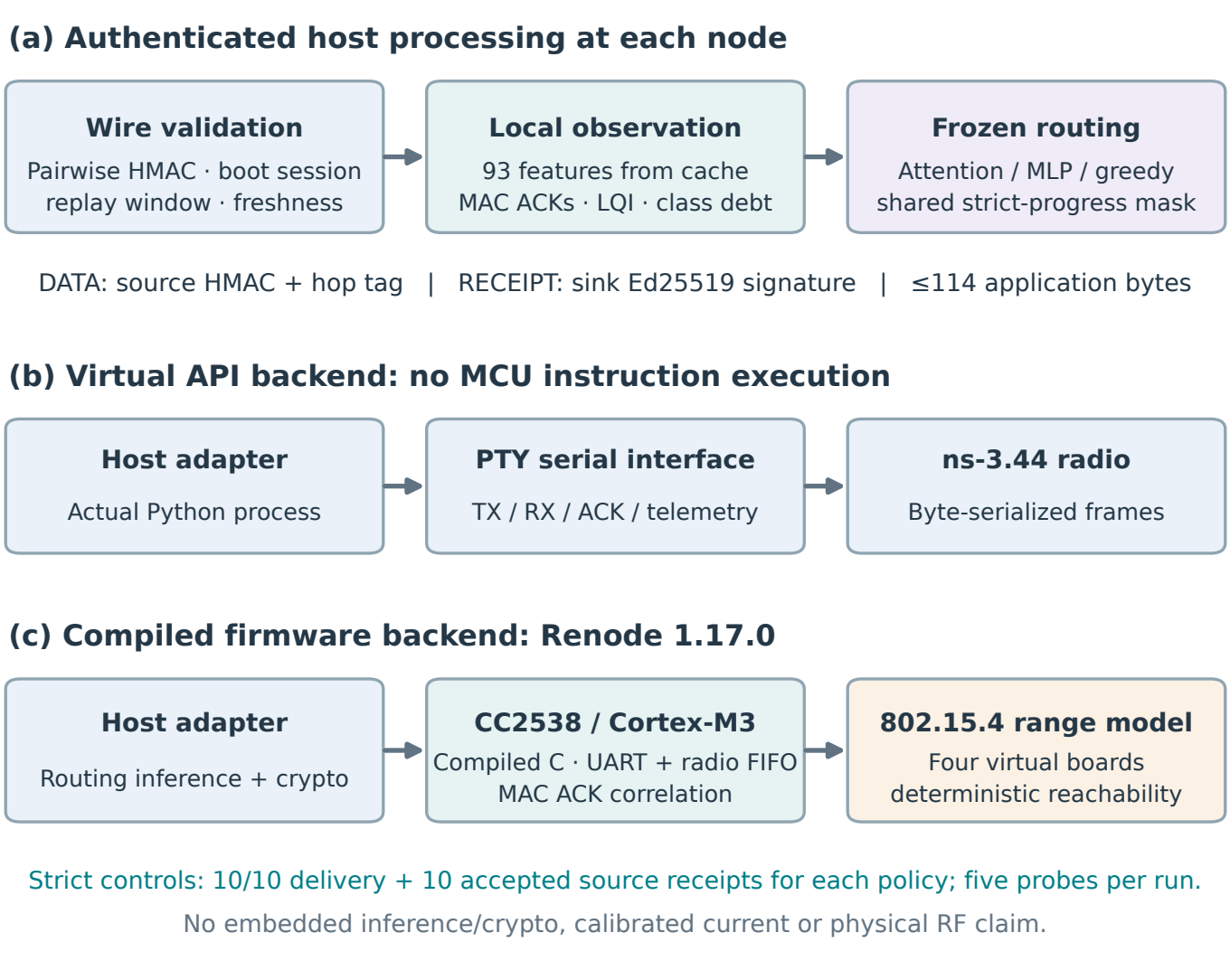
(a) Authenticated host processing at each node
Wire validation
Pairwise HMAC · boot session
replay window · freshness
Local observation
93 features from cache
MAC ACKs · LQI · class debt
Frozen routing
Attention / MLP / greedy
shared strict-progress mask
DATA: source HMAC + hop tag | RECEIPT: sink Ed25519 signature | ≤114 application bytes
(b) Virtual API backend: no MCU instruction execution
Host adapter
Actual Python process
PTY serial interface
TX / RX / ACK / telemetry
ns-3.44 radio
Byte-serialized frames
(c) Compiled firmware backend: Renode 1.17.0
Host adapter
Routing inference + crypto
CC2538 / Cortex-M3
Compiled C · UART + radio FIFO
MAC ACK correlation
802.15.4 range model
Four virtual boards
deterministic reachability
Strict controls: 10/10 delivery + 10 accepted source receipts for each policy; five probes per run.
No embedded inference/crypto, calibrated current or physical RF claim.